

Test 12: Real zeros vs GUE matrix

Verdict: PASS | Generated: 2026-09-02 14:02:26 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Two-sample KS: data vs simulated GUE matrix spectrum.

METHOD: Two-sample KS: data vs simulated GUE matrix spectrum.

Result

Test 12 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

`plots/plot\_01` (.png 600dpi / .svg / .pdf)

Plots

- `plots/animation.gif` ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 12 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(no log_comp entries)

FULL COMPUTATION LOG

??

CONSOLE CAPTURE

TEST 12 (secondary, HARDCORE): Objection 2 (extended): Advanced RMT Diagnostics ? 2KS symmetry audit

??

Full: real n=9600 vs ref n=9600, D=0.023333, p=0.010547 (primary verdict governs)

W1 ref split : D_vs_A=0.028542, D_vs_B=0.020729, |?D|=0.007812 (limit 0.02)

W2 blocks : 4/4 blocks agree with full verdict at ?=0.01 (limit ? 3)

W3 effect : D_full=0.023333 < 2KS critical at ?=0.01: 0.023527? YES

??

W1 ref symmetry PASS ? |?D| = 0.007812

W2 block agree PASS ? 4/4 agree

W3 D bound PASS ? D 0.0233 vs crit 0.0235

Test 12 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS